AI-Generated Art: From Screenplays to Images & Beyond

Surf code： 20230065

Supervisor: Jieming Ma

Team number: Yizhao Chen, Fanyun Yang, Yifei Shen, Machao Yang, Xining Ge, Dongyu Mo, Xiaohan Zhang and Fengze Li

Abstract

Generative models are designed to produce observations stochastically, adhering to specific underlying parameters. Diffusion models, due to their robust generative capabilities, have recently risen to prominence within the realm of generative modeling. Notably, in the field of computer vision, the efficacy of diffusion models has undergone extensive scrutiny. In light of this, the present project endeavors to develop a diffusion model-based AI art effect generator. This generator aims to visualize artistic images and elucidate the conceptualizations behind the text, translating various text inputs into their corresponding video outputs. Moreover, this initiative also seeks to introduce an image-to-video model predicated on diffusion models and adversarial generative networks, further enhancing the capacity for AI-driven art video generation.

Introduction

Diffusion models[1] constitute a subset of deep generative models that have recently emerged as a focal point in the domain of computer vision. As of now, these models have been harnessed for a diverse range of generative modeling endeavors, encompassing image generation[2], image super-resolution, image inpainting, image editing, and image-to-image translation, to list a few.

In the diffusion phase, the continuous introduction of noise transforms the original data, transitioning it from its initial distribution to a desired distribution. For instance, through the persistent addition of Gaussian noise, the original data distribution can be modified to conform to a normal distribution. In the subsequent inverse diffusion phase, a neural network facilitates the restoration of the data, transitioning it from a normal distribution back to its original state.

Within the scope of this project, participants will delve into the mathematical foundations, architectural intricacies, and training methodologies intrinsic to diffusion modeling. The endeavor will also encompass applying the models to intricate planning tasks. Beyond independently training our models, the project entails leveraging open-source AI speech generation models and employing third-party editing tools to refine the resultant audio tracks. Ultimately, our aim is to harness the Advanced Image Generation and Composition (AIGC) to produce any desired image, video, or theatrical scene — such as the simulated face-to-face dialogue between XJTLU's President Xi and Baidu CEO Robin Li, as exemplified in this initiative.

Methodology

The basic idea behind diffusion models is rather simple. They take the input image $\mathbf{x}_0$ and gradually add Gaussian noise to it through a series of T steps. We will call this the forward process. Notably, this part is necessary to generate the targets for our neural network (the image after applying $t < T$ noise steps). Afterward, a neural network is trained to recover the original data by reversing the noising process. By being able to model the reverse process, we can generate new data. This is the so-called reverse diffusion process or, in general, the sampling process of a generative model [3].

Forward diffusion

Diffusion models can be seen as latent variable models. In practice, they are formulated using a Markov chain of T steps. Here, a Markov chain means that each step only depends on the previous one, which is a mild assumption. Importantly, we are not constrained to using a specific type of neural network, unlike flow-based models. Given a data-point $\mathbf{x}_0$ sampled from the real data distribution $q(x)(\mathbf{x}_0 \sim q(x))$, one can define a forward diffusion process by adding noise. Specifically, at each step of the Markov chain we add Gaussian noise with variance β_t to $\mathbf{x}_{t-1}$,, producing a new latent variable $\mathbf{x}_t$ with distribution $q(\mathbf{x}_t \mid \mathbf{x}_{t-1})$. This diffusion process can be formulated as follows:

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}\left(\mathbf{x}_t; \boldsymbol{\mu}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \boldsymbol{\Sigma}_t = \beta_t \mathbf{I}\right). \quad (1)$$

Since we are in the multi-dimensional scenario $\mathbf{I}$ is the identity matrix, indicating that each dimension has the same standard deviation β_t. Note that $q(\mathbf{x}_t \mid \mathbf{x}_{t-1})$ is still a normal distribution, defined by the mean $\boldsymbol{\mu}$ and the variance $\boldsymbol{\Sigma}$ where $\boldsymbol{\mu}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1}$ and $\boldsymbol{\Sigma}_t = \beta_t \mathbf{I}$. $\boldsymbol{\Sigma}$ will always be a diagonal matrix of variances (here β_t). Thus, we can go in a closed form from the input data $\mathbf{x}_0$ to $\mathbf{x}_T$ in a tractable way. Mathematically, this is the posterior probability and is defined as Equation 2:

$$q(\mathbf{x}_{1:T} \mid \mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t \mid \mathbf{x}_{t-1}), \quad (2)$$

where $q(\mathbf{x}_{1:T})$ states that we apply q repeatedly from 1 to T, shown as the left upper part of the Figure 1.

Summary of the principles and training process of the model

Diffusion models operate by iteratively introducing Gaussian noise to the original image across a set of T steps, a technique termed as 'diffusion'. To generate novel data, the reverse diffusion trajectory is emulated using a neural network. The training paradigm for this model hinges on the optimisation of the evidence lower bound (ELBO). One intriguing feature of diffusion models is their ability to be conditioned on variables such as image labels or text embeddings, which in effect "steer" the diffusion trajectory. To render the models compatible with high resolutions, methodologies like Cascade and Latent diffusion have been posited. Cascade diffusion models function in a sequential manner, progressively producing images with enhanced resolutions. Conversely, Latent diffusion models implement the diffusion trajectory within a constrained latent space, ensuring computational efficacy. This is achieved by employing a variational autoencoder for both upscaling and downscaling processes. In parallel, score-based models introduce noise perturbations in a sequential manner to the pristine image.

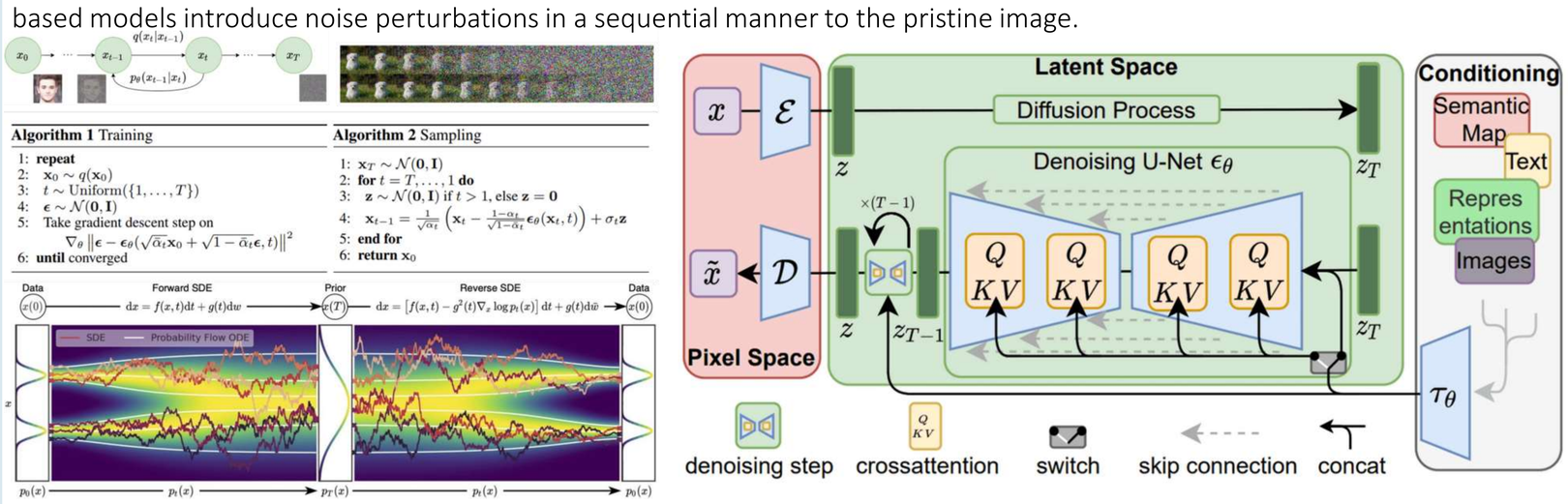

Fig.1. A multifaceted explanation and schematic of how the diffusion model works.

Experiments and Results

Given the heterogeneity in image dimensions, preprocessing is a crucial step before any subsequent operations. The preprocessing entails resizing each image to a standardized dimension defined within config.image_size, augmenting the dataset using the RandomHorizontalFlip method, which introduces variability by randomly mirroring images, and normalizing pixel values to fall within the [-1, 1] range, a requirement of the model. Furthermore, pretrained models, such as those available in Diffusers, can be effortlessly instantiated from their corresponding model classes based on the desired parameters.

Conclusion

In the process of generating images and videos, this project employed a diffusion model-based image generator to craft the appearance of the two avatars representing bishops. To produce the auditory dialogue for these avatars, an extensive dataset of authentic audio samples was curated and refined. Subsequent audio editing and enhancement were facilitated using third-party software, in conjunction with IIElevenLabs, ensuring the avatars' voices were of high fidelity. The script for the culminating video was derived from Baidu's Ernie Bot, a knowledge-enhanced language model termed "Wen Xin Yi Yan LLM." Following meticulous human editing of the script, video generators, specifically Gen-2 and Runway, were employed to finalize the digital human video. This video depicts an engaging discourse between Principal Xi and Robin Li on a pertinent topic.

Acknowledgement

This work was supported by Xian Jiaotong-Liverpool University under Grant SURF20230065.

References

1 J. Sohl-Dickstein, E. Weiss, N. Maheswaranathan and S. Ganguli, "Deep unsupervised learning using non-equilibrium thermodynamics", *Proc. Int. Conf. Mach. Learn.*, pp. 2256-2265, 2015.

2 A. Q. Nichol and P. Dhariwal, "Improved denoising diffusion probabilistic models", *Proc. Int. Conf. Mach. Learn.*, pp. 8162-8171, 2021.

3 Ho, Jonathan, Ajay Jain, and Pieter Abbeel. "Denoising diffusion probabilistic models." Advances in neural information processing systems 33 (2020): 6840-6851.

Video Download
https://pan.baidu.com/s/1FedO0qFxPGssdlc1NLmjpA
PSW: qnnr